

1. General Description

The EM74HC174 and EM74HCT174 are hex positive edge-triggered D-type flip-flops with individual data inputs (D_n) and outputs (Q_n). The common clock (CP) and master reset ($\overline{MR}$) inputs load and reset all flip-flops simultaneously. The D-input that meets the set-up and hold time requirements on the LOW-to-HIGH clock transition is stored in the flip-flop and appears at the Q output. A LOW on $\overline{MR}$ causes the flip-flops and outputs to be reset LOW. Inputs include clamp diodes. This enables the use of current limiting resistors to interface inputs to voltages in excess of V_{CC} .

2. Features and Benefits

- Wide operating voltage 2.0 V to 6.0 V
- High noise immunity
- CMOS low power dissipation
- Latch-up performance exceeds 250 mA
- Input levels:
 - For EM74HC174: CMOS level
 - For EM74HCT174: TTL level
- Six edge-triggered D-type flip-flops
- Asynchronous master reset
- Complies with JEDEC standards:
 - JESD8C (2.7 V to 3.6 V)
 - JESD7A (2.0 V to 6.0 V)
- ESD protection:
 - HBM ANSI/ESDA/JEDEC JS-001 Class 2 exceeds 3500 V
 - CDM ANSI/ESDA/JEDEC JS-002 Class C3 exceeds 2000 V
- Multiple package options
- Specified from -40 °C to +85 °C and from -40 °C to +125 °C

EM74HC174; EM74HCT174

Hex D-type flip-flop with reset; positive-edge trigger

3. Ordering Information

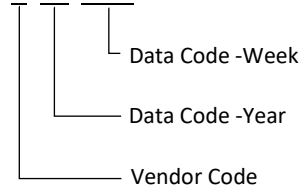
Table 1. Ordering information

Type number	Topside marking	Package		
		Name	Description	Quantity
EM74HC174D	HC174 XYWW	SOP-16L	plastic small outline package; 16 leads; body width 3.9 mm	3000
EM74HCT174D	HCT174 XYWW			
EM74HC174PW	HC174 XYWW	TSSOP16L	plastic thin shrink small outline package; 16 leads; body width 4.4 mm	3000
EM74HCT174PW	HCT174 XYWW			

MARKING INFORMATION

NOTE: XYWW = Vendor Code and Data Code.

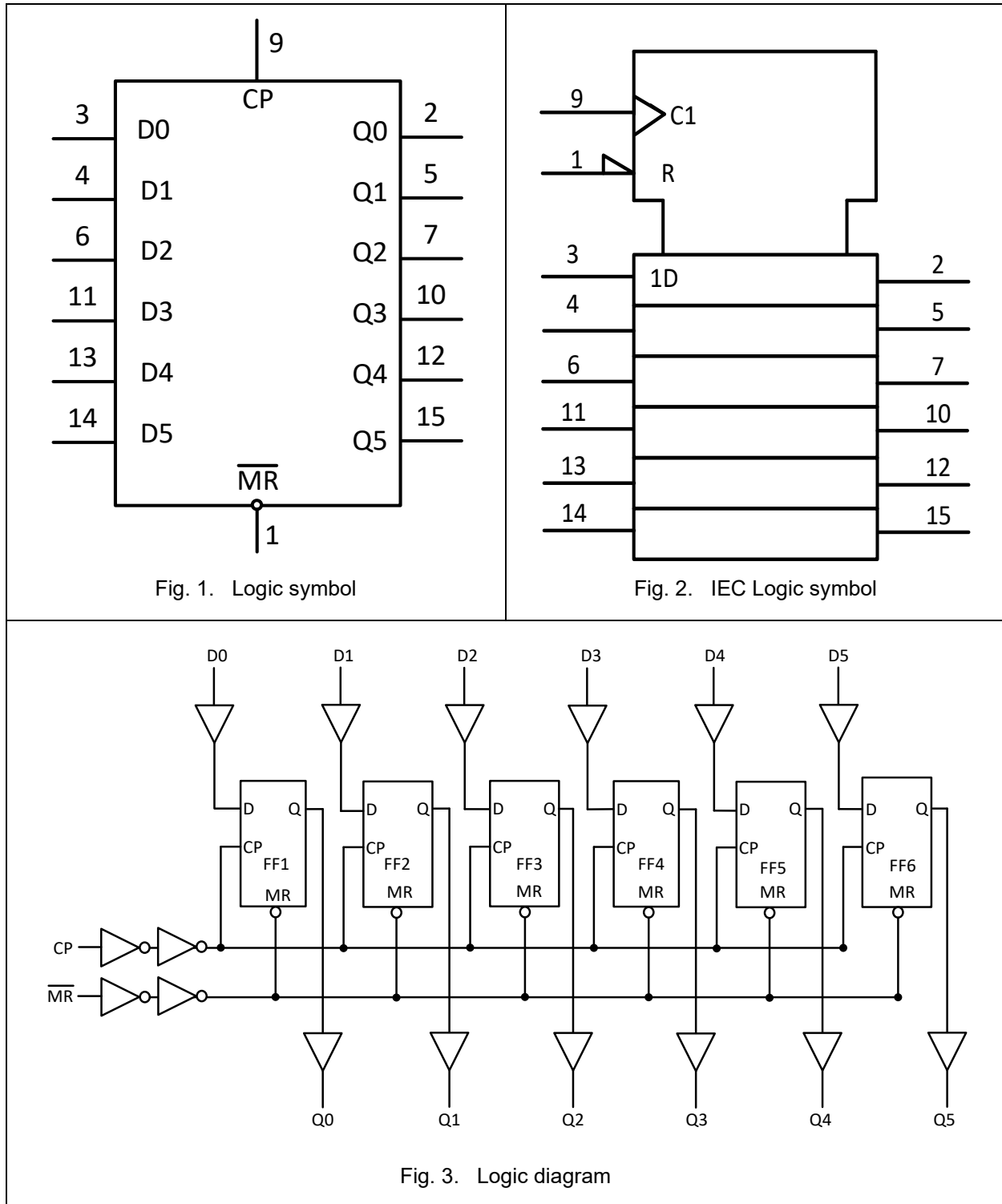
X YY WW



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4. Function Diagram



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5. Pinning Information

5.1. Pinning

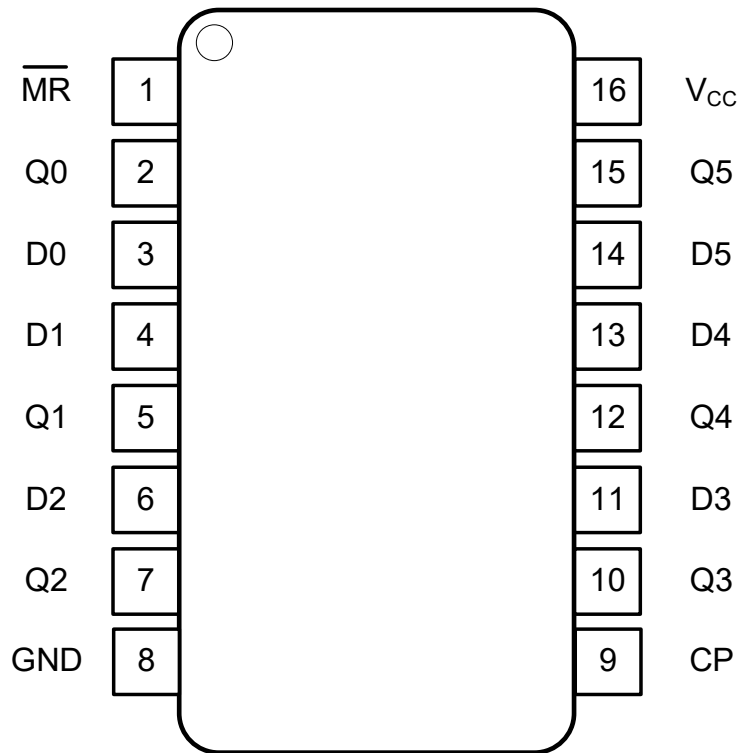


Fig. 4. Top view pin configuration SOP-16L and TSSOP-16L

5.2. Pin description

Table 2. Pin description

Symbol	Pin	Description
$\overline{\text{MR}}$	1	asynchronous master reset input (active LOW)
Q0, Q1, Q2, Q3, Q4, Q5	2, 5, 7, 10, 12, 15	flip-flop output
D0, D1, D2, D3, D4, D5	3, 4, 6, 11, 13, 14	data input
GND	8	ground (0 V)
CP	9	clock input (LOW-to-HIGH edge-triggered)
V _{CC}	16	supply voltage

EM74HC174; EM74HCT174**Hex D-type flip-flop with reset; positive-edge trigger**

6.Functional Description

Table 3. Function table

H = HIGH voltage level; h = HIGH voltage level one set-up time prior to the LOW-to-HIGH clock transition;
L = LOW voltage level; l = LOW voltage level one set-up time prior to the LOW-to-HIGH clock transition;
X = don't care; $\uparrow$ = LOW-to-HIGH clock transition.

Operating modes	Inputs			Outputs
	$\overline{\text{MR}}$	CP	Dn	Qn
reset (clear)	L	X	X	L
load "1"	H	$\uparrow$	h	H
load "0"	H	$\uparrow$	l	L

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7. Absolute Maximum Ratings

Stresses exceeding the absolute maximum ratings may damage the device. The device may not function or be operable above the recommended operating conditions and stressing the parts to these levels is not recommended. In addition, extended exposure to stresses above the recommended operating conditions may affect device reliability. The absolute maximum ratings are stress ratings only.

Table 5. Absolute Maximum Ratings

In accordance with the Absolute Maximum Rating System (IEC 60134). Voltages are referenced to GND.

Symbol	Parameter	Conditions	Min	Max	Unit
V_{CC}	supply voltage		-0.5	7	V
I_{IK}	input clamping current	$V_I < -0.5\text{ V}$ or $V_I > V_{CC} + 0.5\text{ V}$		± 20	mA
I_{OK}	output clamping current	$V_O < -0.5\text{ V}$ or $V_O > V_{CC} + 0.5\text{ V}$		± 20	mA
I_O	output current	$-0.5\text{ V} < V_O < V_{CC} + 0.5\text{ V}$		± 35	mA
I_{CC}	supply current			70	mA
I_{GND}	ground current		-70		mA
P_{tot}	total power dissipation	$T_{amb} = -40\text{ }^{\circ}\text{C}$ to $+125\text{ }^{\circ}\text{C}$		500	mW
T_{stg}	storage temperature		-65	+150	$^{\circ}\text{C}$

8. Recommended Operating Conditions

The Recommended Operating Conditions table defines the conditions for actual device operation. Recommended operating conditions are specified to ensure optimal performance to the datasheet specifications. EnergyMath does not recommend exceeding them or designing to Absolute Maximum Ratings.

Table 6. Recommended Operating Conditions

Symbol	Parameter	Conditions	EM74HC174			EM74HCT174			Unit
			Min	Typ	Max	Min	Typ	Max	
V_{CC}	supply voltage		2.0	5.0	6.0	4.5	5.0	5.5	V
V_I	input voltage		0		V_{CC}	0		V_{CC}	V
V_O	output voltage		0		V_{CC}	0		V_{CC}	V
T_{amb}	ambient temperature		-40	+25	+125	-40	+25	+125	$^{\circ}\text{C}$
$\Delta t/\Delta V$	input transition rise and fall rate	$V_{CC} = 2.0\text{ V}$			625				ns/V
		$V_{CC} = 4.5\text{ V}$		1.67	139		1.67	139	ns/V
		$V_{CC} = 6.0\text{ V}$			83				ns/V

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9.Static Characteristics

Table 6. Static characteristics

At recommended operating conditions. Voltages are referenced to GND (ground = 0 V).

Symbol	Parameter	Conditions	-40 °C to +85 °C			-40 °C to +125 °C		Unit
			Min	Typ[1]	Max	Min	Max	
EM74HC174								
V _{IH}	HIGH-level input voltage	V _{CC} = 2.0 V	1.5			1.5		V
		V _{CC} = 4.5 V	3.15			3.15		V
		V _{CC} = 6.0 V	4.2			4.2		V
V _{IL}	LOW-level input voltage	V _{CC} = 2.0 V			0.5		0.5	V
		V _{CC} = 4.5 V			1.35		1.35	V
		V _{CC} = 6.0 V			1.8		1.8	V
V _{OH}	HIGH-level output voltage	V _I = V _{IH} or V _{IL}						
		I _O = -20μA; V _{CC} = 2.0 V	1.9			1.9		V
		I _O = -20μA; V _{CC} = 4.5 V	4.4			4.4		V
		I _O = -20μA; V _{CC} = 6.0 V	5.9			5.9		V
		I _O = -4.0 mA; V _{CC} = 4.5 V	3.84			3.7		V
		I _O = -5.2 mA; V _{CC} = 6.0 V	5.34			5.2		V
V _{OL}	LOW-level output voltage	V _I = V _{IH} or V _{IL}						
		I _O = 20μA; V _{CC} = 2.0 V			0.1		0.1	V
		I _O = 20μA; V _{CC} = 4.5 V			0.1		0.1	V
		I _O = 20μA; V _{CC} = 6.0 V			0.1		0.1	V
		I _O = 4.0 mA; V _{CC} = 4.5 V			0.33		0.4	V
		I _O = 5.2 mA; V _{CC} = 6.0 V			0.33		0.4	V
I _I	input leakage current	V _I = V _{CC} or GND ; V _{CC} = 6.0 V			±1		±1	μA
I _{CC}	supply current	V _I = V _{CC} or GND ; I _O = 0A ; V _{CC} = 6.0 V			20		40	μA
C _I	input capacitance			3.5				pF

[1]All typical values are measured at $T_{amb} = 25^\circ\text{C}$.

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Hex D-type flip-flop with reset; positive-edge trigger

Symbol	Parameter	Conditions	-40 °C to +85 °C			-40 °C to +125 °C		Unit
			Min	Typ[1]	Max	Min	Max	
EM74HCT174								
V _{IH}	HIGH-level input voltage	V _{CC} = 4.5 V to 5.5 V	2.0			2.0		V
V _{IL}	LOW-level input voltage	V _{CC} = 4.5 V to 5.5 V			0.8		0.8	V
V _{OH}	HIGH-level output voltage	V _I = V _{IH} or V _{IL}						
		I _O = -20µA; V _{CC} = 4.5 V	4.4		4.4		V	
		I _O = -4.0 mA; V _{CC} = 4.5 V	3.84		3.7		V	
V _{OL}	LOW-level output voltage	V _I = V _{IH} or V _{IL}						
		I _O = 20µA; V _{CC} = 4.5 V			0.1		0.1	V
		I _O = 4.0 mA; V _{CC} = 4.5 V			0.33		0.4	V
I _I	input leakage current	V _I = V _{CC} or GND ; V _{CC} = 5.5 V			±1		±1	µA
I _{CC}	supply current	V _I = V _{CC} or GND ; I _O = 0 A ; V _{CC} = 5.5 V			20		40	µA
ΔI _{CC}	additional supply current	per pin ; V _I = V _{CC} - 2.1 V; I _O = 0 A; other inputs at V _{CC} or GND; V _{CC} = 4.5 V to 5.5 V			450		490	µA
C _I	input capacitance			3.5				pF

[1] All typical values are measured at $T_{amb} = 25^\circ\text{C}$.

EM74HC174; EM74HCT174

Hex D-type flip-flop with reset; positive-edge trigger

10. Dynamic Characteristics

Table 7. Dynamic characteristics

Voltages are referenced to GND (ground = 0 V); for test circuit see Fig. 7.

Symbol	Parameter	Conditions	-40 °C to +85 °C			-40 °C to +125 °C		Unit
			Min	Typ[1]	Max	Min	Max	
EM74HC174								
t _{pd}	propagation delay	CP to Qn; see Fig. 5 [2]						
		V _{CC} = 2.0 V			55		60	ns
		V _{CC} = 4.5 V			20		25	ns
		V _{CC} = 6.0 V			15		18	ns
t _{PHL}	HIGH to LOW propagation delay	$\overline{MR}$ to Qn; see Fig. 6						
		V _{CC} = 2.0 V			190		225	ns
		V _{CC} = 4.5 V			38		45	ns
		V _{CC} = 6.0 V			33		38	ns
t _t	transition time	Qn; see Fig. 5 [3]						
		V _{CC} = 2.0 V			95		110	ns
		V _{CC} = 4.5 V			19		22	ns
		V _{CC} = 6.0 V			16		19	ns
t _w	pulse width	CP HIGH or LOW; see Fig. 5						
		V _{CC} = 2.0 V	100			120		ns
		V _{CC} = 4.5 V	20			24		ns
		V _{CC} = 6.0 V	17			20		ns
		$\overline{MR}$ input LOW; see Fig.6						
		V _{CC} = 2.0 V	100			120		ns
		V _{CC} = 4.5 V	20			24		ns
		V _{CC} = 6.0 V	17			20		ns
t _{rec}	recovery time	$\overline{MR}$ to CP; see Fig.6						
		V _{CC} = 2.0 V	5			5		ns
		V _{CC} = 4.5 V	5			5		ns
		V _{CC} = 6.0 V	5			5		ns
t _{su}	set up time	Dn to CP; see Fig.5						
		V _{CC} = 2.0 V	75			90		ns
		V _{CC} = 4.5 V	15			18		ns
		V _{CC} = 6.0 V	13			15		ns

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Hex D-type flip-flop with reset; positive-edge trigger

Symbol	Parameter	Conditions	-40 °C to +85 °C			-40 °C to +125 °C		Unit
			Min	Typ[1]	Max	Min	Max	
t_h	hold time	Dn to CP; see Fig.5						
		$V_{CC} = 2.0 \text{ V}$	3			3		ns
		$V_{CC} = 4.5 \text{ V}$	3			3		ns
		$V_{CC} = 6.0 \text{ V}$	3			3		ns
f_{max}	maximum frequency	CP input; see Fig.5						
		$V_{CC} = 2.0 \text{ V}$	5			4		MHz
		$V_{CC} = 4.5 \text{ V}$	24			20		MHz
		$V_{CC} = 6.0 \text{ V}$	28			24		MHz
C_{PD}	power dissipation capacitance	$f_i = 1 \text{ MHz};$ $V_I = \text{GND to } V_{CC};$ [4]		35				pF
EM74HCT174								
t_{pd}	propagation delay	CP to Qn; see Fig. 5 [2]						
		$V_{CC} = 4.5 \text{ V}$			20		25	ns
t_{PHL}	HIGH to LOW propagation delay	$\overline{MR}$ to Qn; see Fig. 6						
		$V_{CC} = 4.5 \text{ V}$	44			53		ns
t_t	transition time	Qn; see Fig. 5 [3]						
		$V_{CC} = 4.5 \text{ V}$			19		22	ns
t_w	pulse width	CP input; see Fig. 5						
		$V_{CC} = 4.5 \text{ V}$	20			24		ns
		$\overline{MR}$ input LOW; see Fig.6						
		$V_{CC} = 4.5 \text{ V}$	25			30		ns
t_{rec}	recovery time	$\overline{MR}$ to CP; see Fig.6;						
		$V_{CC} = 4.5 \text{ V}$	15			18		ns
t_{su}	set up time	Dn to CP; see Fig.5						
		$V_{CC} = 4.5 \text{ V}$	20			24		ns
t_h	hold time	Dn to CP; see Fig.5						
		$V_{CC} = 4.5 \text{ V}$	5			5		ns
f_{max}	maximum frequency	CP input; see Fig.5						
		$V_{CC} = 4.5 \text{ V}$	24			20		MHz
C_{PD}	power dissipation capacitance	$f_i = 1 \text{ MHz};$ $V_I = \text{GND to } V_{CC} - 1.5 \text{ V};$ [4]		41				pF

EM74HC174; EM74HCT174

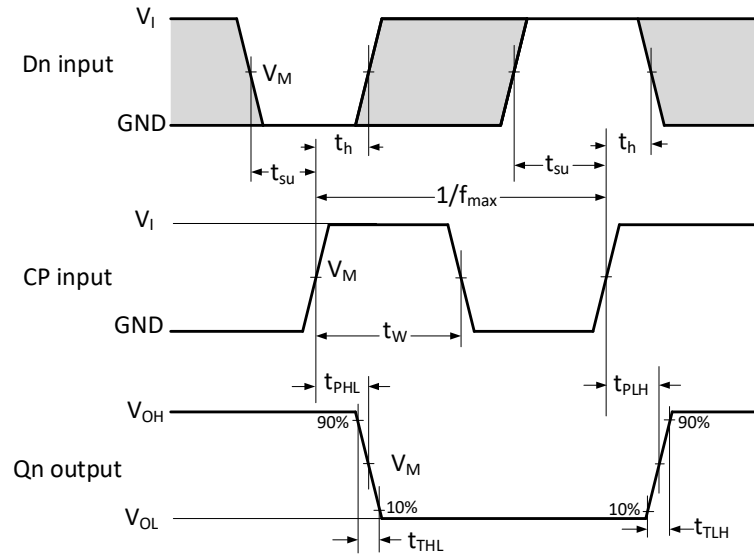
Hex D-type flip-flop with reset; positive-edge trigger

- [1] Typical values are measured at $T_{amb} = 25\text{ }^{\circ}\text{C}$.
- [2] t_{pd} is the same as t_{PLH} and t_{PHL} .
- [3] t_t is the same as t_{THL} and t_{TLH} .
- [4] C_{PD} is used to determine the dynamic power dissipation (P_D in μW).
- $P_D = C_{PD} \times V_{CC}^2 \times f_i \times N + \Sigma(C_L \times V_{CC}^2 \times f_o)$ where:
- f_i = input frequency in MHz;
 - f_o = output frequency in MHz;
 - C_L = output load capacitance in pF;
 - V_{CC} = supply voltage in V;
 - N = number of inputs switching;
 - $\Sigma(C_L \times V_{CC}^2 \times f_o)$ = sum of outputs.

EM74HC174; EM74HCT174

Hex D-type flip-flop with reset; positive-edge trigger

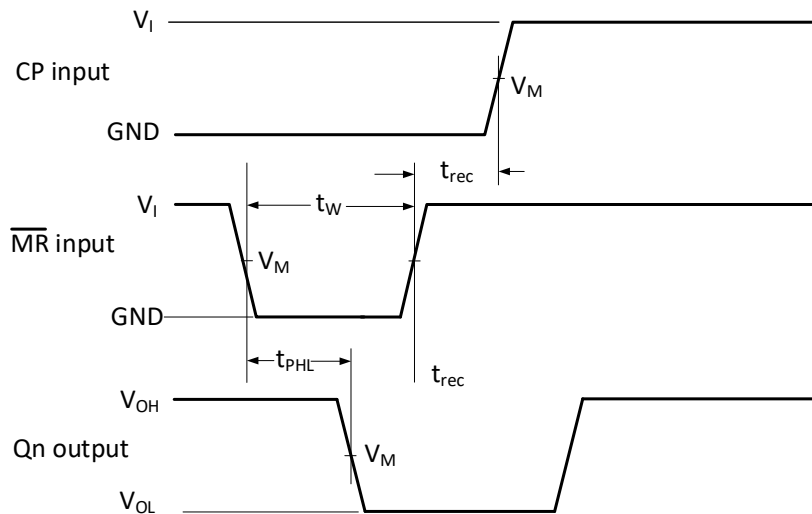
10.1. Waveforms and test circuit



Measurement points are given in Table 8.

The shaded areas indicate when the input is permitted to change for predictable output performance. V_{OL} and V_{OH} are typical voltage output levels that occur with the output load.

Fig. 5. Input to output propagation delay, output transition time, clock input pulse width, set-up and hold times for data input and maximum frequency



Measurement points are given in Table 8.

V_{OL} and V_{OH} are typical voltage output levels that occur with the output load.

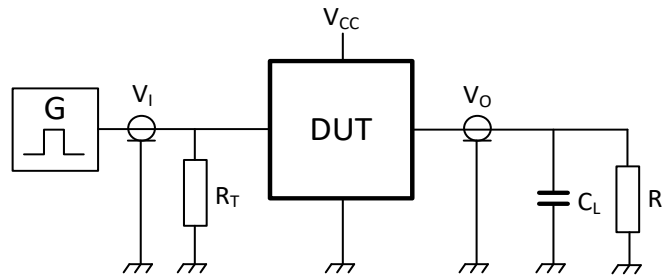
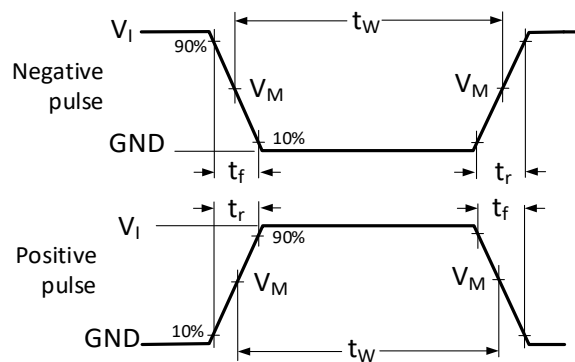
Fig. 6. Master reset to output propagation delays, master reset pulse width and master reset to clock recovery time

EM74HC174; EM74HCT174

Hex D-type flip-flop with reset; positive-edge trigger

Table 8. Measurement points

Type	Input		Output
	V_I	V_M	V_M
EM74HC174	V_{CC}	$0.5V_{CC}$	$0.5V_{CC}$
EM74HCT174	3 V	1.3 V	1.3 V



Test data is given in Table 9.

Definitions for test circuit:

R_L = Load resistance.

C_L = Load capacitance including jig and probe capacitance.

R_T = Termination resistance should be equal to the output impedance Z_o of the pulse generator.

V_{EXT} = External voltage for measuring switching times.

Fig. 7. Test circuit for measuring switching times

Table 9. Test data

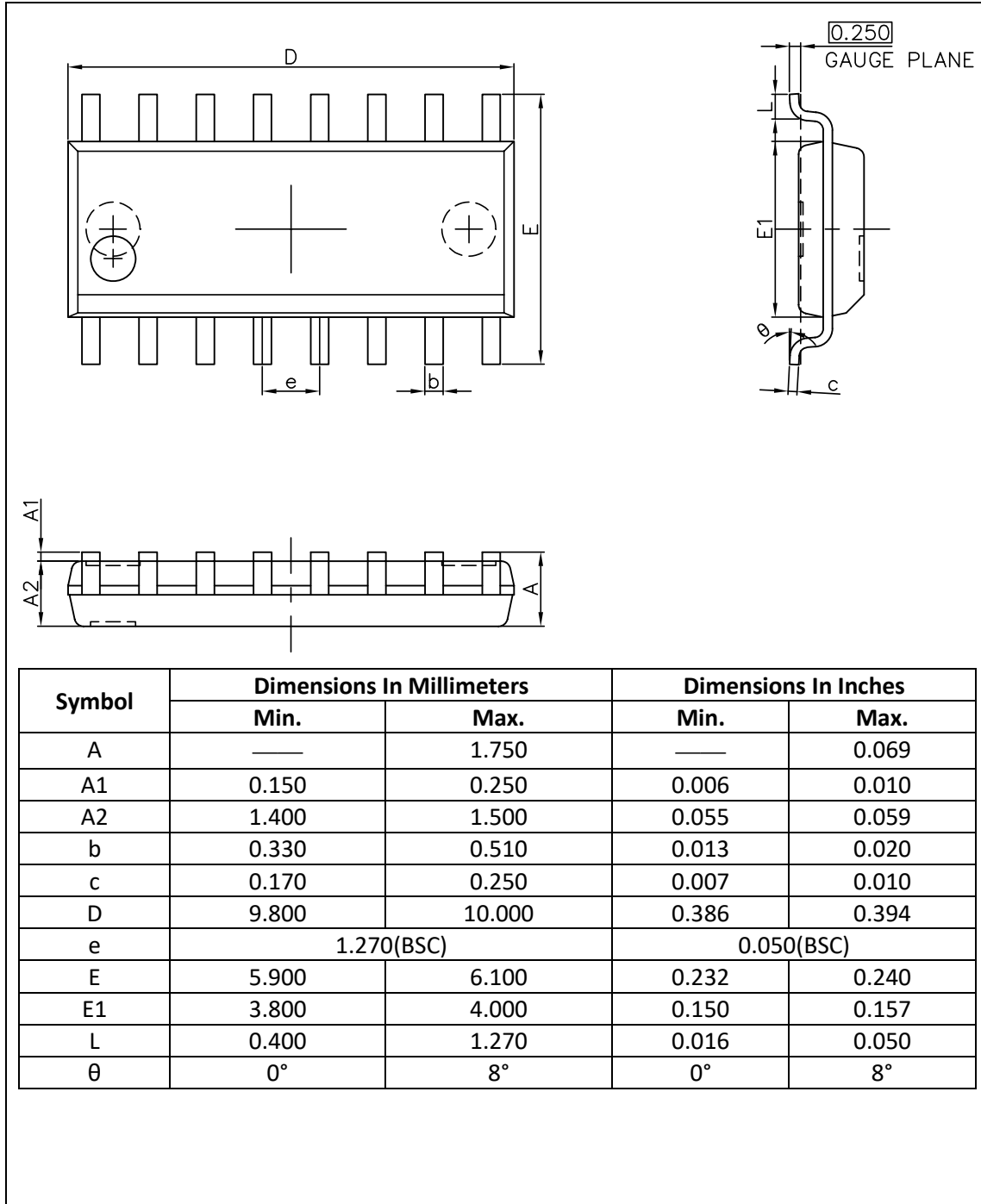
Type	Input		Load	
	V_I	$t_r = t_f$	C_L	R_L
EM74HC174	V_{CC}	2.5 ns	15 pF	500 Ω
EM74HCT174	3 V	2.5 ns	15 pF	500 Ω

EM74HC174; EM74HCT174

Hex D-type flip-flop with reset; positive-edge trigger

11. Package Outline

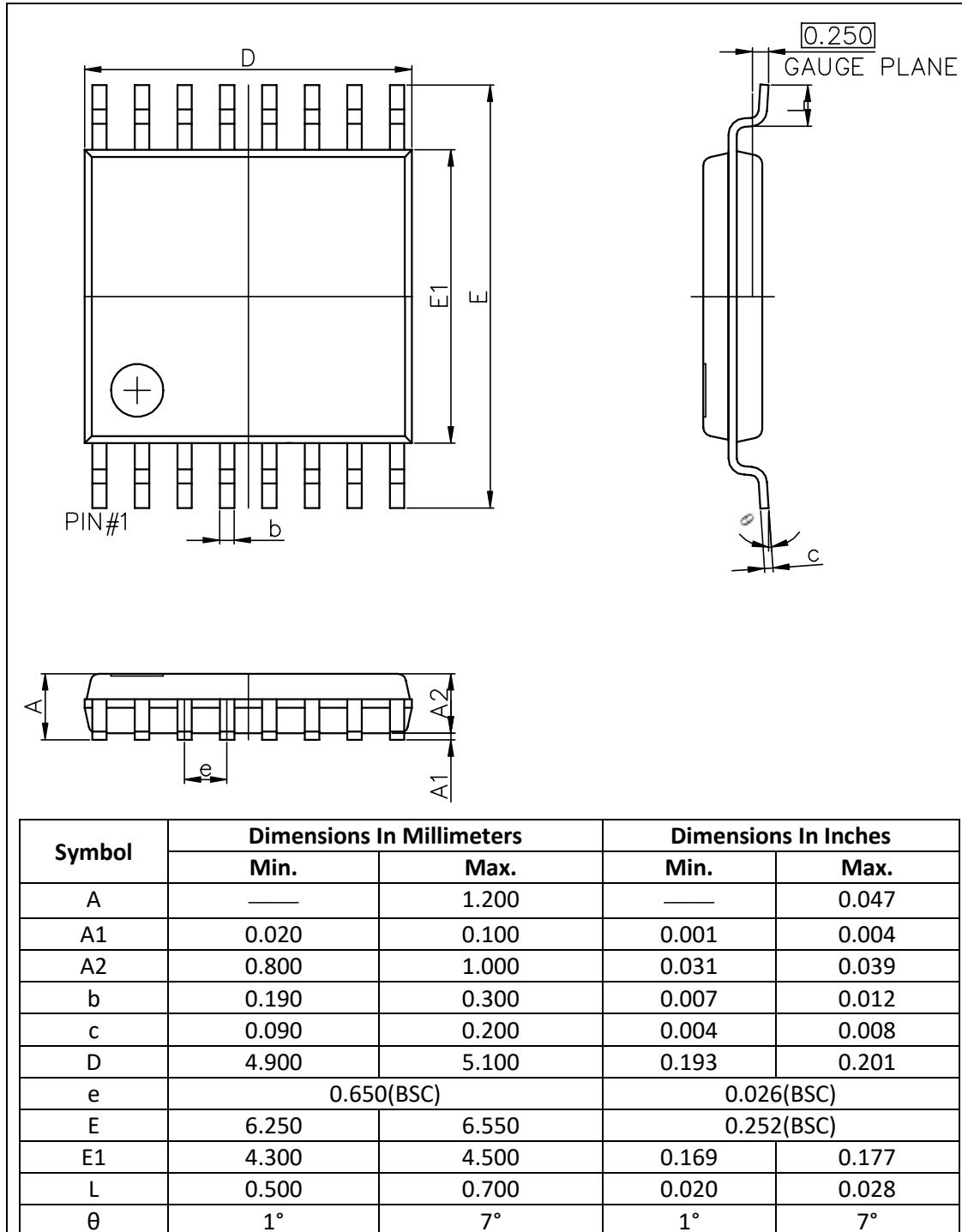
SOP-16L



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TSSOP-16L



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12. Tape and Reel Information

12.1. Carrier tape dimensions

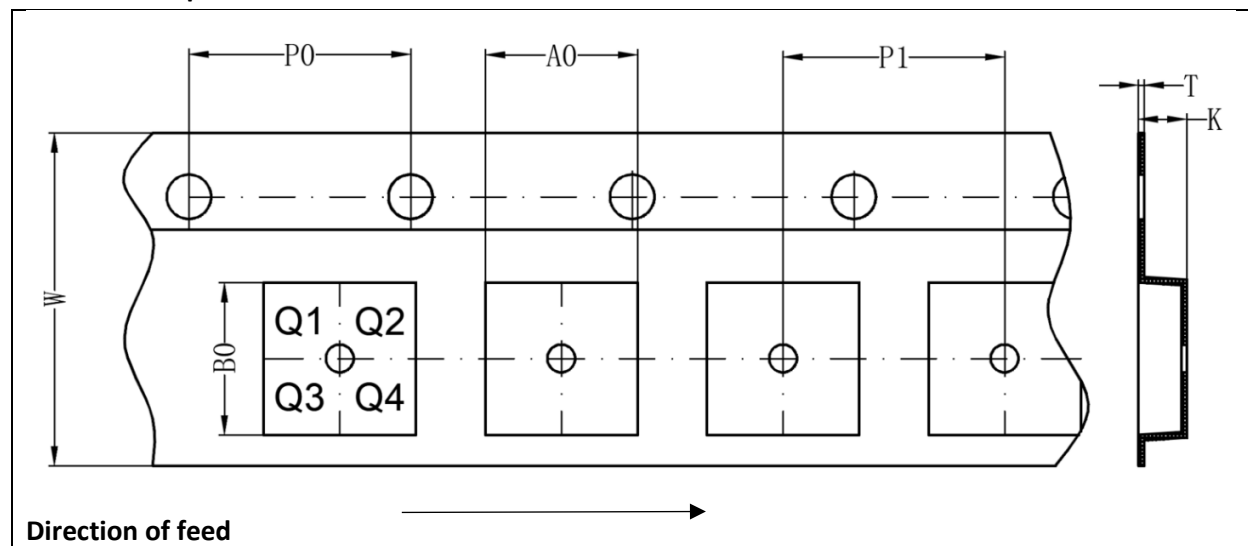


Table 10. Carrier tape dimensions

Package version	A0(mm)	B0(mm)	K0(mm)	T(mm)	P1(mm)	W(mm)	P0(mm)	PIN 1
SOP-16L	6.5	10.45	2.1	0.22	8	16	4	Q1
TSSOP-16L	6.7	5.45	1.6	0.25	8	12	4	Q1

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12.2.Reel and box dimensions

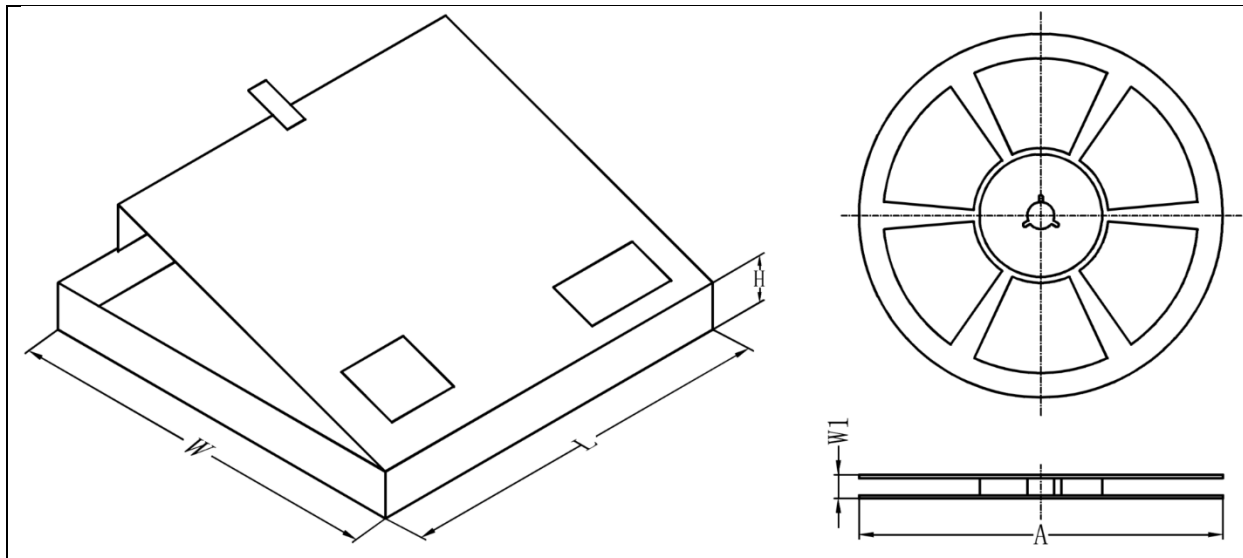


Table 11. Dimensions and quantities

Package version	Type NO. ending	Reel Dimension A (mm)	Reel Width W1 (mm)	SPQ (pcs)[1]	Reels per box	Outer box dimensions L×W×H(mm) [2]
SOP-16L	D	330	22.4	3000	1	358x340x50
TSSOP-16L	PW	330	18.4	3000	1	358x340x50

[1] Packing quantity dependent on specific product type. Please contact your local Energymath representative for ordering.

[2] Dimensions for reference only.

EM74HC174; EM74HCT174

Hex D-type flip-flop with reset; positive-edge trigger

13. Abbreviations

Table 12. Abbreviations

Acronym	Description
CMOS	Complementary Metal-Oxide Semiconductor
DUT	Device Under Test
ESD	ElectroStatic Discharge
HBM	Human Body Model
CDM	Charged Device Model
TTL	Transistor-Transistor Logic

14. Revision History

Table 13. Revision history

Document ID	Release Date	Data sheet status	Change notice	Supersedes
EM74HC_HCT174 Rev. 1.0	May 20, 2025	Product datasheet		